

Lab C Terraform Success Log

After multiple iterations and troubleshooting sessions, the lab was successfully completed. Here's the breakdown of what it took to get a working Terraform-deployed EC2 → RDS Notes App.

The Path to Success

Phase 1: Infrastructure Deployment

- Terraform successfully created VPC, subnets, NAT, IGW, RDS, EC2, Security Groups, Secrets Manager, SSM Parameters, CloudWatch resources
- Multiple `terraform apply` cycles required due to resource conflicts from Lab 1A/1B (see Errors page)

Phase 2: App Accessibility

- **Problem:** Website not loading externally despite app running locally
- **Root Cause:** Port mismatch — Flask app running on port 5000, but Security Group only allowed port 80
- **Fix:** Added port 5000 to EC2 Security Group inbound rules
- **Verification:** `curl localhost:5000/health` returned healthy, then external access worked

Phase 3: RDS Connectivity (`/init` failing)

- **Problem:** `OperationalError: Can't connect to MySQL server on 'lab-mysql...' (Name or service not known)`
- **Root Cause 1:** Secret name mismatch
 - Terraform created secret: `helga/rds/mysql`
 - App was looking for: `lab/rds/mysql` (from old `user_data.sh`)
- **Root Cause 2:** Wrong host in secret
 - Secret had: `lab-mysql.chkce02amfxr.us-east-2.rds.amazonaws.com` (old RDS)

- Should have been: `helga-rds01.chkce02amfxr.us-east-2.rds.amazonaws.com` (current RDS)
- **Root Cause 3:** User data only runs once
 - Updated `user_data.sh` file doesn't affect already-running EC2 instances
 - Must terminate and relaunch EC2 for new `user_data` to take effect

Phase 4: Final Fix

1. Verified the correct RDS endpoint: `helga-rds01.chkce02amfxr.us-east-2.rds.amazonaws.com`
2. Updated the Secrets Manager secret with correct host
3. Changed the password in the secret to match the new RDS instance
4. Terminated and relaunched EC2 instance to pick up corrected `user_data.sh`
5. `/init` succeeded — database and table created

Key Lessons Learned



When debugging, go back to basics.

Multiple script versions caused confusion. The fix was to identify the simplest working version, verify each component independently, and layer complexity back in one piece at a time.



User data only runs once.

Editing `user_data.sh` in Terraform does NOT update running EC2 instances. You must terminate and relaunch (or SSH in and fix manually).



Secrets Manager + Terraform naming matters.

If your Terraform uses `${ local.name _prefix }/rds/mysql` for the secret name, your app must reference that exact name (`helga/rds/mysql`), not a hardcoded default like `lab/rds/mysql`.

Final Working Configuration

Component	Value
Secret Name	helga/rds/mysql
RDS Endpoint	helga-rds01.chkce02amfxr.us-east-2.rds.amazonaws.com
Database Name	t_labdb
App Port	80
EC2 SG Inbound	Port 80 (HTTP), Port 22 (SSH)
RDS SG Inbound	Port 3306 from EC2 SG

Verification Commands Used

```
# Check app is running
systemctl status rdsapp

# Test locally
curl localhost/health
curl localhost/init
curl "localhost/add?note=test"
curl localhost/list

# Check secret contents
aws secretsmanager get-secret-value --secret-id helga/rds/mysql --query "SecretString" --output text | jq .

# Check RDS endpoint
aws rds describe-db-instances --query "DBInstances[].Endpoint.Address" --output text
```